

DeepCSAT – Ecommerce Customer Satisfaction Score Prediction

Project Report

1. Introduction

Customer Satisfaction (CSAT) is an important metric for evaluating how well a company's products and services meet customer expectations. In the e-commerce industry, analyzing customer satisfaction helps organizations identify service gaps, improve customer experience, and increase retention.

This project focuses on predicting Customer Satisfaction (CSAT) scores for an e-commerce platform using machine learning and deep learning techniques. By analyzing customer interaction data, the system predicts whether a customer is satisfied or not and identifies the key factors influencing satisfaction.

2. Objective of the Project

The main objectives of this project are:

1. To analyze customer interaction data from an e-commerce platform.
 2. To preprocess and clean the dataset for machine learning models.
 3. To build predictive models for customer satisfaction.
 4. To compare different machine learning and deep learning models.
 5. To identify important features affecting customer satisfaction.
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3. Dataset Description

The dataset contains information about customer interactions with the e-commerce platform.

Important features include:

- Customer interaction details
- Response time
- Customer support behavior
- Product related information
- Service quality indicators

The target variable is **Customer Satisfaction Score (CSAT)** which indicates whether the customer experience was satisfactory.

4. Data Preprocessing

Before training the models, several preprocessing steps were performed:

4.1 Handling Missing Values

Missing values were handled using appropriate techniques such as filling with mean/median or removing invalid records.

4.2 Feature Engineering

New features were created and categorical variables were converted into numerical format using encoding techniques.

4.3 Data Scaling

Standardization was applied using **StandardScaler** to normalize the feature values and improve model performance.

4.4 Train-Test Split

The dataset was divided into:

- **Training data**
- **Testing data**

This helps evaluate how well the model performs on unseen data.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was performed to understand the dataset.

Key tasks included:

- Distribution analysis of features
- Correlation analysis
- Outlier detection
- Visualization of important variables

EDA helped identify patterns and relationships between features and the CSAT score.

6. Machine Learning Models Used

Multiple machine learning and deep learning models were implemented:

6.1 Random Forest

Random Forest is an ensemble learning algorithm that builds multiple decision trees and combines their predictions to improve accuracy and reduce overfitting.

6.2 XGBoost

XGBoost is a gradient boosting algorithm that improves model performance by sequentially reducing prediction errors.

6.3 Artificial Neural Network (ANN)

A deep learning model was built using neural networks to capture complex patterns in the dataset.

The ANN model includes:

- Input layer
- Hidden layers
- Output layer

Activation functions and optimization techniques were used to improve learning efficiency.

7. Model Evaluation

The performance of the models was evaluated using several metrics:

Evaluation Metrics

- **Precision**
Measures how many predicted positive cases were actually correct.
- **Recall**
Measures how many actual positive cases were correctly predicted.
- **F1 Score**
Harmonic mean of Precision and Recall.
- **ROC-AUC Curve**
Measures the model's ability to distinguish between classes.

These metrics help determine which model performs best.

8. Feature Importance Analysis

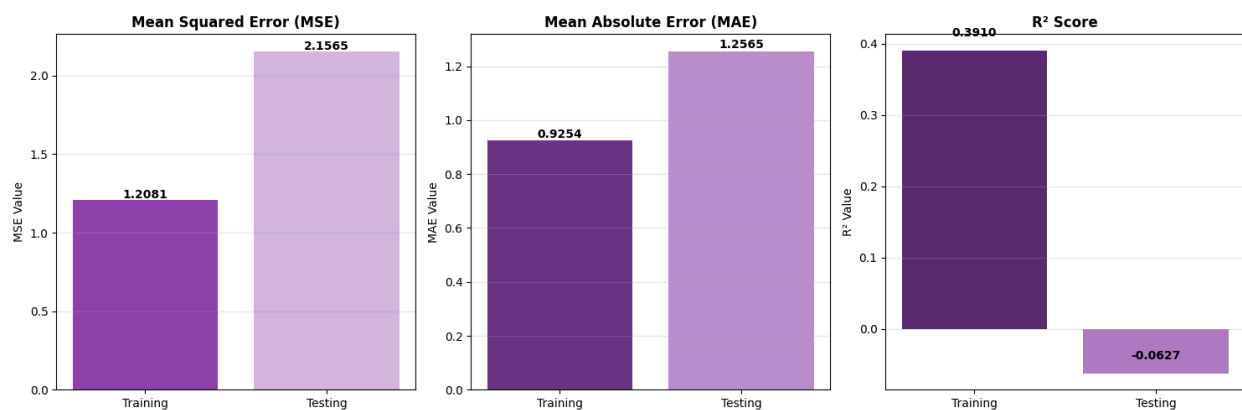
Feature importance analysis was performed to determine which factors most influence customer satisfaction.

Important factors may include:

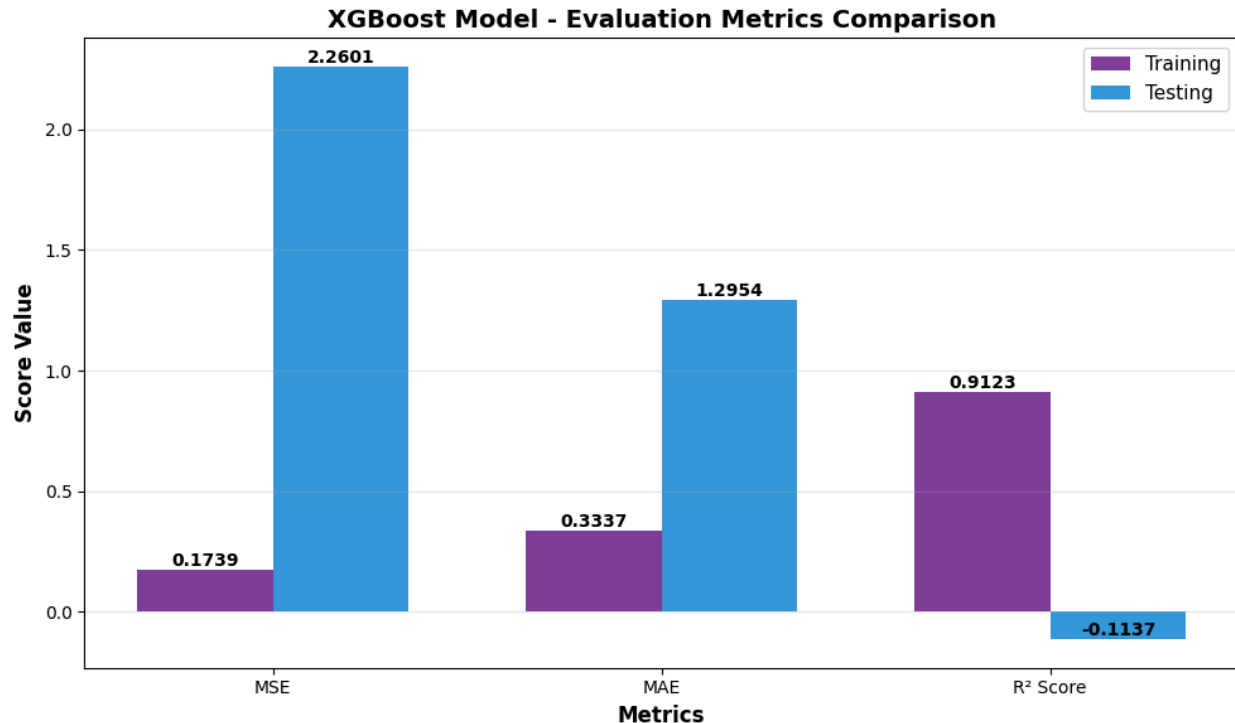
- Customer support response time
- Issue resolution quality
- Product experience
- Interaction quality

Understanding these features helps businesses improve service quality.

Explain the ML Model used and it's performance using Evaluation metric Score Chart ANN.



Explain the ML Model XG BOOST used and it's performance using Evaluation metric Score Chart



9. Results and Findings

Key insights from the project:

1. Machine learning models can effectively predict customer satisfaction.
2. Ensemble models like Random Forest and XGBoost perform well on structured data.
3. Deep learning models capture complex relationships in customer behavior.
4. Feature importance analysis highlights critical service factors affecting satisfaction.

10. Conclusion

This project successfully demonstrates how machine learning and deep learning techniques can be used to predict customer satisfaction in the e-commerce domain.

By analyzing customer interaction data and training predictive models, businesses can identify dissatisfaction factors and improve service quality. The insights generated from this model can help companies enhance customer experience and build stronger relationships with users.

11. Future Improvements

Future enhancements could include:

- Using larger datasets for better model accuracy
- Implementing advanced deep learning architectures

12. Deploying the model using Streamlit

Prediction Results

Predicted CSAT Score

3.13/5.00

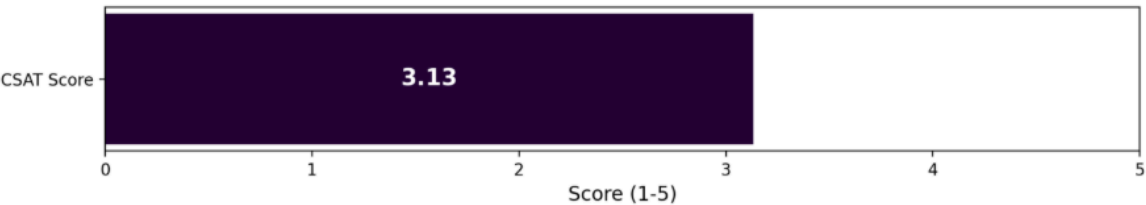
Satisfaction Level

62.7%

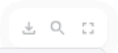
Customer Sentiment

😊 Satisfied

Satisfaction Gauge



Input Features Summary



	Feature	Value	Description
0	Response_Time	15	Response Time (minutes to first response)
1	Resolution_Time	8	Resolution Time (hours to complete resolution)
2	Agent_Professionalism	7.5	Agent Professionalism Score (1-10)
3	Communication_Quality	7.5	Communication Quality Score (1-10)
4	Issue_Complexity	5	Issue Complexity Level (1-10)
5	Follow_up_Quality	7.5	Follow-up Quality Score (1-10)
6	First_Contact_Resolution	75	First Contact Resolution Rate (%)
7	Customer_Effort_Score	6	Customer Effort Score (1-10)